Towed-Sensor Surveillance UAV

AMERICAN UNIVERSITY OF THE MIDDLE EAST

1.0 EXECUTIVE SUMMARY

Objective. Machine Masters will design, build and fly a hand-flown electric aircraft that carries, tows and protects the 2027 sensor payload, to post the highest total score at Tucson in April 2027.

Approach. The rules became 21 numbered subsystem requirements. A mission model calibrated on the 2022–26 fly-offs flew 162 candidate aircraft through every mission, failed attempts included, against simulated rival fields. Completion pays 4.5 of the 7.5 mission points and the sensitivity study gives reliability 55 % of the trade weight; declaring the sensor mass at technical inspection is worth +0.42 points.

Design. A 1.82 m, 1.00 m² high wing with Fowler flaps ($C_{L,\max}$ 1.93), conventional tail, fixed tricycle gear and one 8S tractor; at its 14.6 kg Mission 2 mass it stalls at 11.9 m/s. It waives Mission 1, flies one 7.4 kg unit, tows 3.4 kg and drops the declared sensor from 1.524 m: 6.76 of 7.5 with every mission completed and, counting failed attempts, a 0.313 chance of the field's top mission total (our first iteration: 0.128). A slotted 1.2 m² wing is the dated fallback.

Execution. Six sub-teams build two airframes, retire the top risks on the bench by 15 January and freeze the design on 19 February; the $38.0k budget, travel included, is university funded.

DESIGN AT A GLANCE / VARIANT A

WINGSPAN	WING AREA	M2 GROSS MASS	STALL, M2	SCORE, IF DONE	P(TOP OF FIELD)
1.82 m	**1.00** m²	**14.6** kg	**11.9** m/s	**6.76** / 7.5	**0.313**

2.0 MANAGEMENT SUMMARY

2.1 Team Organization

Machine Masters has 21 members in six sub-teams organized by function, mirroring how the aircraft is scored (Fig. 02, Table 01). The faculty advisor signs off spending and flight-safety reviews; the team lead owns schedule, budget and rules compliance; the chief engineer owns the configuration and alone may change a requirement. Sensor Systems exists because the sensor, drum and container touch three of the four mission scores. Sub-teams meet weekly and review mass, schedule and budget from one version-controlled repository. Each decision records its evidence and the test that would overturn it, and is reviewed at PDR (28 Nov), the A/B gate with CDR (15 Jan) and configuration freeze (19 Feb). The pilot is an AMA member.

2.2 Schedule and Milestones

The schedule (Fig. 01) is built backward from the fly-off. Parts common to both variants are ordered first, so the A/B decision costs one wing, never a second program. The prototype, with the Variant A wing, is built in December and flies on 6 January; its weigh-in and lap times recalibrate the model before the design report. The competition airframe is built from 16 January and flies on 14 February.

2.3 Budget

Table 02 covers the year: the purchase list, a second airframe and backup wing, 8 kg of tungsten at the $66/kg online average, and for 7 travellers the quoted fares ($1,545 each), meals at the $80/day federal rate and $435 visa fees.[1] A 15 % contingency covers a crash re-

[1] Lastminute.com quotes for 12–20 Apr. 2027; GSA FY2027 Tucson per diem; U.S. visa fees; six tungsten-alloy listings (accio.ai, IndiaMART); read 31 Oct. 2026.

TABLE 01 Sub-teams, what each owns and the skills each requires. Every scored subsystem has exactly one owner.

SUB-TEAM	OWNS	REQUIRED SKILLS
Aero & Performance	Sizing, lap model, stability, CFD	Python modelling; XFoil, AVL, OpenFOAM
Structures & Manufacturing	Wing, Fowler tracks, fuselage, tests	CAD; laser cutting, printing, foam cutting
Propulsion & Avionics	Motor, propeller, pack, fuse, radio	Thrust-stand testing; LiPo safety; ExpressLRS
Sensor Systems	Dart, lights, drum, container, test box	Embedded C; servo drives; cushioning design
Flight Operations	Pilot, observer, crew, checklists	AMA-member pilot; timed staging drills
Business & Logistics	Budget, purchasing, travel, visas	Lead-time planning; cost control; writing

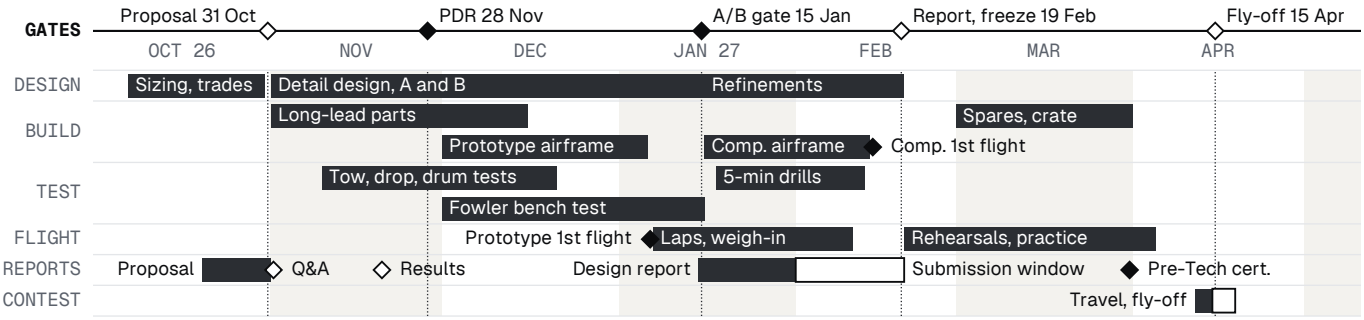


FIG. 01 Decision-gated schedule: long-lead parts first; the prototype flies the Variant A wing in early January; the Fowler test decides A or B on 15 Jan; the configuration freezes with the report on 19 Feb. Outlined: dates fixed by the rules; filled: team plan.

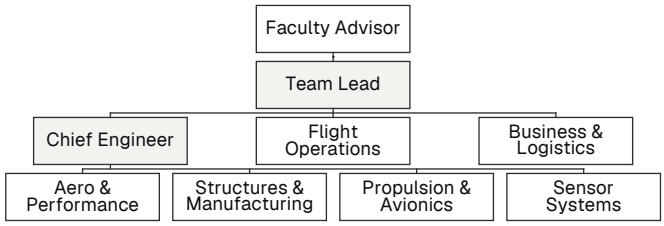


FIG. 02 One owner for every scored subsystem; Sensor Systems is a peer of the airframe teams.

TABLE 02 Program budget for the year, every expected cost including travel (USD, rows rounded to $10). List: itemized purchase list at vendor prices; quote: retrieved 31 Oct 2026; avg.: online average; GSA: US per diem; est.: estimate.

ITEM	BASIS	COST
Airframes, B wing, Fowler test, rigs	list, est.	3,260
Propulsion, radio, servos, 8S packs	list	4,800
Sensor, drum, containers, tungsten	list, avg.	1,320
Props, adhesives, consumables	list	460
Crate; Tucson paint, glue	list, est.	250
Airfare, Kuwait–Tucson, 7 people	quote	10,820
Lodging, 4 rooms, 6 nights	quote	2,850
Meals, 7 people, 6.5 days	GSA	3,640
Van, fuel, crate baggage	est.	1,200
US visa fees, 7 people	fee	3,050
Fly-off fee; AIAA, AMA dues	rules, fee	1,420
Contingency, 15 %		4,960
Total		**38,030**
Funding: university, full cost	confirmed	38,030

build. The university funds the full cost: 10,000 KWD ($32,170) is allocated now and the $5,860 balance is drawn on the same approval.

3.0 CONCEPTUAL DESIGN APPROACH

3.1 Mission Requirements Decomposition

Each mission score and rule became a numbered subsystem requirement (SR, Table 03); the SR column of Table 09 shows the test that verifies each.

3.2 Scoring Model and Sensitivity Study

The mission score sums four terms, each a completion credit plus a ratio to the best result in the field, our own included:[2]

$$S_{\mathrm{M}} = M_1 + \left[1 + \frac{(W/t)}{(W/t)_{\max}}\right] + \left[2 + \frac{NW_s}{(NW_s)_{\max}}\right] + \left[0.5 + \frac{W_s h}{(W_s h)_{\max}}\right] \le 7.5 \tag{1}$$

Completion is worth 4.5 points and the bonuses at most 3.0, yet in 2026 only 17 of the 83 teams that passed technical inspection completed every mission.[3] The declared sensor mass W_s sets the Ground Mission product, every Mission 2 unit and the Mission 3 ballast cap. Winning mission totals ran 5.61–6.80 in 2022–26.

Each design is flown 120 times with every uncertain input drawn (mass, $C_{L,\max}$, drag, efficiency, density and a

[2] AIAA, *2026–27 Design, Build, Fly Rules*, 2 Sep. 2026, Secs. 3.1–3.3, 6.3.

[3] AIAA, "DBF Competition Summaries," 2022–2026 (513 team results).

TABLE 03 Mission requirements decomposed into subsystem requirements (SR). † Team requirement beyond the rules.

MISSION	SCORE	KEY CONSTRAINTS	DERIVED SUBSYSTEM REQUIREMENTS
M1 Staging	$M_1 = 1.0$	3 laps in 5 min; may be waived and credited by a successful M2	SR-01 static margin 16 % MAC in every loading†; SR-02 all checks inside the 5-min staging
M2 Delivery	$M_2 = 1 + \frac{W/t}{(W/t)_{\max}}$	Sensor in container plus simulators, each at the declared maximum; TOP up; 5 laps	SR-03 bay for a 288 × 303 × 303 mm container; SR-04 positive restraint; SR-05 CG inside marked limits in every loading
M3 Sensor	$M_3 = 2 + \frac{NW_s}{(NW_s)_{\max}}$	Internal until deployed; deploy before the first downwind turn; OFF, SOLID, FLASHING, OFF; stow before landing	SR-06 permanent internal drum, tow ≥ 2.73 m, latch holds the stowed sensor; SR-07 three lights, modes over the line, visible in sun; SR-08 no divergent swing at n = 3.24; SR-09 own COTS battery; SR-10 line clear of propeller, tail†; SR-11 same sensor-door state in every mission
Ground	$GM = 0.5 + \frac{W_s h}{(W_s h)_{\max}}$	6 drops on concrete, $h \leq$ 1.524 m (60 in); ring to bottom < 381 mm (15 in); 5-min rigging	SR-12 container absorbs $W_s h$, sensor rated to 150 g peak deceleration; SR-13 tool-less O/E rigging, flush fittings; SR-14 test box drives all three modes
System	–	Span ≤ 1.829 m (6 ft); TOGW < 24.95 kg (55 lb); ≤ 100 Wh; no autopilot, GPS or camera; arming, fuse, radio rules	SR-15 pilot-in-the-loop only; SR-16 positive latches, no magnets; SR-17 §4.2.5 fail-safe, range check; SR-18 structure for n = 3.24 and tip lift; SR-19 top-surface arming plug ≥ 152 mm behind the propeller plane, external Rx switch, separate Rx pack, BEC off, blade fuse ≤ pack rating; SR-20 M2 at full payload within 78.1 Wh usable, stall ≤ 12 m/s; SR-21 ESC, pack within rated temperature†

lap-time factor from 24 flights past teams measured[4]), declares its sensor at inspection and meets 5,000 simulated rival fields, with each attempt's success drawn from 2023–26 results. In the sensitivity study (Fig. 03) efficiency, airframe mass and span efficiency lead the design parameters (0.22, 0.16 and 0.13 points per 10 %; usable energy acts as efficiency), and wing area is at its optimum (±10 % moves the mean under 0.03). Lap time dominates the conditions: 10 % slower laps cost 0.54 points. Both races are energy-bound (77.9 of 78.1 Wh in Mission 2). The largest lever is the declaration (Fig. 04): frozen at design, the best is 6.33 (4 kg); declared at inspection, the same aircraft averages 6.76 with every mission completed.

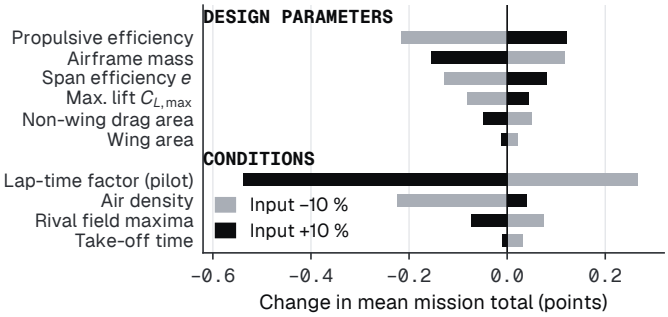


FIG. 03 Sensitivity of the mean mission total to each input, ±10 %, on 120 paired draws about Variant A. Lap time and density, which the team does not set, outweigh every design parameter.

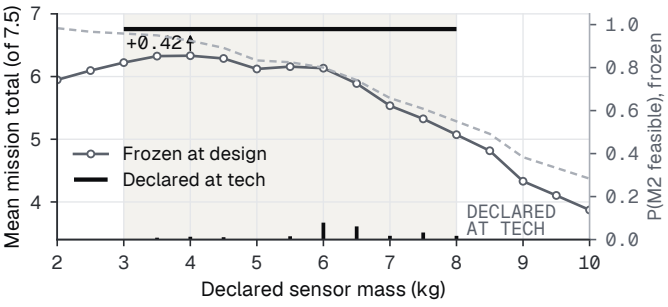


FIG. 04 Declaring at inspection is worth +0.42 over the best frozen mass; frozen at 8 kg, Mission 2 is feasible on only 0.55 of airframes. Shaded: the 3–8 kg window; bars: masses the 120 draws declare.

3.3 Configuration Trade Studies

Trade weights come from the sensitivity study: completion 55 %, lap performance 17 %, high lift 10 %, airframe mass 7 %, efficiency 5 %, induced drag 4 %, structure and parasite drag 1 % each. Of 162 screened combinations of wing area, flap, declared range and propulsion, the eleven best were re-flown on all 120 draws (Fig. 05) and had to pass two gates: Mission 3 completion within 10 % of the best, and a worst-tenth total no lower than the first iteration's. The 1.0 m² Fowler wing

[4] Predicted and measured times in seven DBF design reports, 2021–26; airframe masses of 14 top-three aircraft, 2022–26.

TABLE 04 Configuration decisions, alternatives and evidence. Points: change in mean mission total; matrix: weighted decision matrix, of 100.

DECISION	SELECTED	ALTERNATIVES, EVIDENCE
Planform	**Straight**	Delta −1.69, biplane −0.32
Flap	**Fowler, 1.0**	P(top) 0.012–0.033 lower
Airfoil	**SD7037**	Best of 13; SD7062 −25 % tripped
Wing	**High, 90**	Low 71, mid 53 (matrix)
Tail	**Conv., 83**	T 73, twin boom 68; V fails §4.2.5
Gear	**Trike, 85**	Taildragger 76, bicycle 51
Power	**One 8S**	12S −0.02, twin −0.05 in P(top)
Tips	**Plain**	Winglets −0.01 to −0.05
Sensor	**At tech**	Frozen −0.42 (Fig. 04)

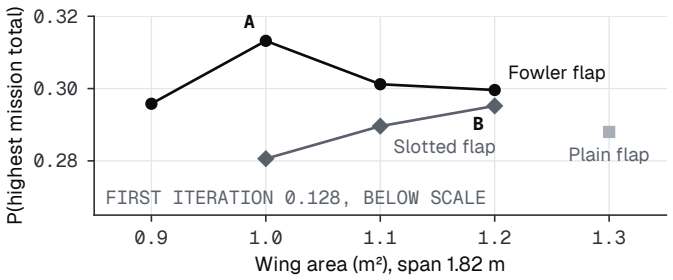


FIG. 05 Finalists re-flown on 120 paired draws: the 1.0 m² Fowler wing (A) leads; the slotted 1.2 m² wing (B) is the fallback. One 8S motor, best declared range per point.

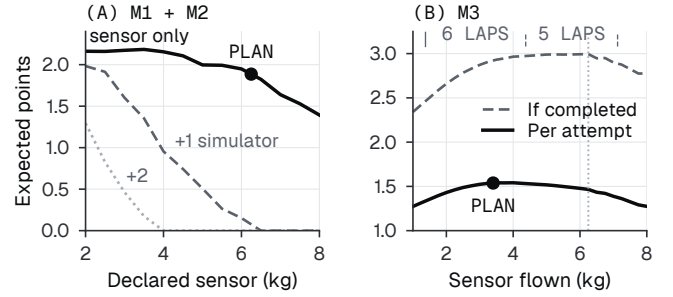


FIG. 06 Payload trades. (a) A simulator never pays: its mass costs more completion than its W/t earns (one at 6 kg declared is feasible in 0.06 of draws). (b) Above about 3.3 kg the lap·kg product reaches the expected field best (20.2), so more ballast adds little if completed and costs completion; per attempt the optimum is flat at 3.25–4.25 kg. Dotted: 6.25 kg declared.

leads every other finalist by 0.012–0.033 in P(top), each paired 95 % interval above zero; larger Fowler wings land slower but are heavier and draggier every lap.[5] Table 04, Fig. 06 and Table 05 give the other decisions, the payload trades and the drum trade.

TABLE 05 Deployment-and-recovery trade (scores 1–5; totals of the scores shown). Boom, wireless link, bridle, pulley and spring reel failed the rules or the 20 s deploy limit.

CRITERION	WT.	CLOSED DRUM	OPEN DRUM	FREE SPOOL
Recovery reliability	0.75	5.0	4.4	3.6
Manufacturability	0.15	3.1	5.0	1.7
Weight, volume	0.04	5.0	4.9	2.4
Signal path	0.03	5.0	5.0	5.0
Deploy, stow time	0.03	4.5	5.0	5.0
Weighted total	**1.00**	**4.70**	**4.55**	**3.35**

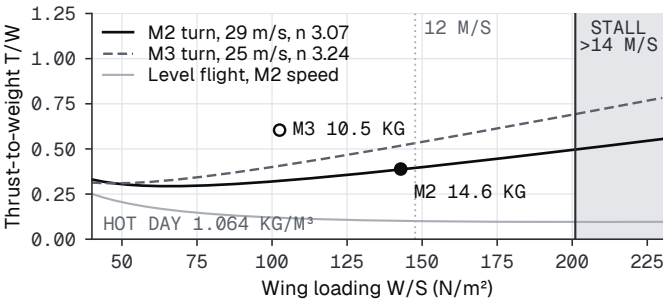


FIG. 07 Mission 2 is sized by the race turn at full thrust; the Fowler flap keeps its 14.6 kg landing under 12 m/s.

3.4 Preliminary Design and Sizing

3.4.1 Sizing method Sizing uses the hot-day density[6] of 1.064 kg/m³ and a lap model of the 152.4 m (500 ft) legs, 180° turns and downwind 360°, optimizing speed and bank under stall (≤ 14 m/s), turn lift, load factor (≤ 4), battery power and the 100 Wh cap, with the measured ×1.15 lap factor. In Fig. 07 the Mission 2 point sits on its 29 m/s, n = 3.07 turn line at full thrust; Mission 3 is limited by energy. From 1,575 motor, propeller and pack combinations, the AT7215 with a 20×15E on 8S gives 101 N static thrust and averages 1.4 kW (47 A) in Mission 2.[7]

3.4.2 Aircraft concept Variant A puts a straight wing of aspect ratio 3.31 (Fig. 08, Table 06) on a 0.345 × 0.355 m box fuselage, with a conventional tail on a 1.05 m arm. Bay and tow exit sit on the CG, so every loading flies at 16 % static margin; about the CG, AVL gives $C_{m\alpha}$ −0.61, $C_{n\beta}$ 0.076 and $C_{l\beta}$ −0.012 per rad.[8] The empty mass is a fit to 14 top-three aircraft, an

[5] Flaps: R. C. Platt, NACA Rept. 534, 1935; C. J. Wenzinger and T. A. Harris, NACA Rept. 664, 1939; method validated on NACA TN 663. Airfoils: M. S. Selig et al., UIUC Low-Speed Airfoil Tests.

[6] 10th percentile at the daily high, 2,878 hourly KTUS observations, 15–18 Apr. 1996–2025, reduced to TIMPA (667 m). Iowa Environmental Mesonet, "ASOS Data Download," retrieved 11 Oct. 2026.

[7] T-Motor and APC published test data; the model is within 3.5 % of a measured 2023 Mission 2 power.

[8] M. Drela and H. Youngren, AVL 3.52; M. Drela, XFoil 6.99; MIT.

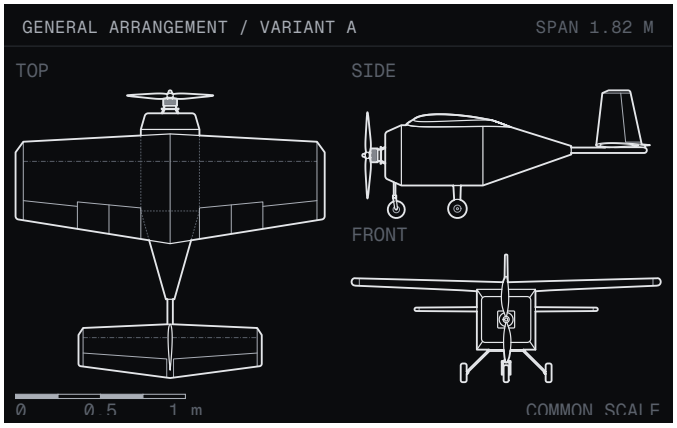


FIG. 08 Variant A to scale from the AVL geometry: a 1.00 m² Fowler-flap high wing (SD7037) over a box fuselage sized around the container; the payload bay and the tow exit sit on the centre of gravity, so every loading flies the same.

TABLE 06 Preliminary design parameters, Variant A. Gross masses and stall speeds are medians of 120 draws, so they do not add to empty mass plus payload.

PARAMETER	VALUE	UNIT
Span / area / aspect ratio	1.82 / 1.00 / 3.31	m, m²
Mean chord / tail arm	0.555 / 1.05	m
Tail areas, H / V	0.291 / 0.069	m²
Empty mass with pack (fit)	6.96	kg
Gross mass, M2 / M3	14.6 / 10.5	kg
Trimmed landing $C_{L,\max}$	1.93	–
Stall, M2 / M3	11.9 / 10.1	m/s
C_{D0} / L/D, M2 level	0.0216 / 9.8	–
Static margin / $C_{m\alpha}$	16 / −0.61	%, /rad
Static thrust / 8S pack	101 / 97.7	N, Wh
M2 mean power / current	1.4 / 47	kW, A
Blade fuse, 58 V rated	80	A
Tow line, built / minimum	2.83 / 2.73	m

estimate until the first weigh-in. Variant B changes only the wing (1.2 m², slotted), tail and boom.

3.4.3 **Sensor and D&R system** The sensor is a dart with a tungsten core, about 200 × 70 mm at 6 kg, rated to 150 g peak deceleration and towed from an eye above its CG so it flies level.[9] Its own 1S pack drives three LED arrays of about 220 cd. A servo-driven closed-loop drum with a slip ring carries the line and the receiver's mode signal. It deploys 3.4 kg in about 10 s, before the first downwind turn, and after the last lap stows it in 16.5 s (24.6 s at 8 kg) into a latched dock that holds it for take-off and landing. In a 3.24 g turn the line carries about 107 N at 3.4 kg and 254 N at 8 kg; line and anchor take four times the 3 g load. The aircraft is flown pilot-in-the-loop: no autopilot, only optional rate gyros.

TABLE 07 Mission targets and scores, Variant A. Targets: medians of 120 draws (P10–P90). Scores: means against the expected field, each mission completed.

MISSION	TARGET	SCORE
M1	Waived; credited by the first M2	1.00
M2	7.4 kg unit; 231 s (158–287)	1.46
M3	3.4 kg; 20.7 lap·kg (16.6–25.8)	2.85
GM	6.25 kg from 1.524 m: 9.53 kg·m	1.44
Total if completed, mean (P10)		**6.76 (6.25)**
Expected, failed attempts counted		**4.83**
P(highest mission total)		**0.313**

3.4.4 **Container and rigging** Built once for the top of the declared range, the container's 41 mm of polyethylene foam keeps the sensor under its 150 g rating in a 1.524 m drop, with the ring under 381 mm from every face.[10] Pads are re-cut at inspection for the declared mass, simulators carry ballast adjustable within 28 g, rigging uses flush, tool-less O/E fittings, and every hatch latches mechanically, never with magnets.

3.5 Mission Design Approach and Scoring Targets

Every plan protects completion first (Table 07); speeds and lap times are the design point. **M1:** waived and credited by Mission 2; at the field's typical 65 % success per attempt and three attempts, this lifts the chance of finishing Mission 3 from 0.40 to 0.72 (Variant A: 0.82 for M2, 0.54 for M3). **M2:** one unit, no simulators (Fig. 06a), at 29 m/s with 71° turns: 38.9 s laps, 204 s with take-off. **M3:** deploy with lights OFF upwind, SOLID after lap 1, FLASHING after lap 2, OFF and stow after the last lap. Ballast is chosen per attempt (Fig. 06b): a lighter dart stalls slower and climbs harder, so 3.4 kg flies 6 laps of 39.9 s at 25 m/s. Charging the stow to the window, as the model does, costs only 0.009 points. **GM:** 6.25 kg from 1.524 m gives 9.53 kg·m, above the expected field best (9.14) in most draws; rigging drilled under 240 s. **Switch rule:** if the Fowler section gives trimmed $C_{L,\max}$ below 1.75, or jams more than once in 100 cycles, by

[9] J. Barrowman, "Theoretical Prediction of the Center of Pressure," NARAM-8, 1966.

[10] U.S. Dept. of Defense, *MIL-HDBK-304B, Package Cushioning Design*, 1978.

15 January, Variant B is built; it trails A by 0.018 in P(top) when the flap works and leads by 0.044 when it does not.

4.0 MANUFACTURING PLAN

4.1 Preliminary Manufacturing Flow

Airframe, sensor and Ground Mission lanes run in parallel and meet at integration (Fig. 09). The prototype is built in five weeks in December and the competition airframe in four from 16 January, from the same files and jigs; the wing splits in two for the airline crate. Built-up structure on pultruded carbon tubes was chosen over moulded composites, whose moulds and cure time the schedule cannot carry; every part is weighed against its mass budget.

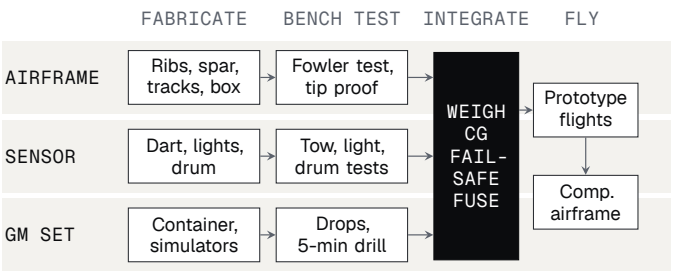


FIG. 09 Airframe, sensor and Ground Mission lanes are each proven on the bench before they meet at integration.

TABLE 08 Components and critical processes: material, method and proof.

COMPONENT	MATERIAL, METHOD	PROOF
Wing	Ply ribs, carbon spar	Tip-lift load
Fowler flaps	Printed carriage, CNC rail	100 cycles
Fuselage, tail	Ply, carbon boom, balsa	Weigh-in
Gear	Aluminium, faired nose leg	Landings
Drum	Printed drum, servo drive	100 cycles
Dart	Potted tungsten core	Tow, drops
Container	CNC-cut PE foam, ply box	Six drops
Harness	10 AWG, 58 V blade fuse	Fuse duty

4.2 Critical Processes and Technologies

The Fowler track is the one technology the team has never built, so a jigged test section is made first and chooses the variant (Table 08). The harness uses a Littelfuse MAXI 58V 0999080.ZXN, an 80 A slow-blow blade fuse rated for 58 V, in a 58 V holder; the common 100 A blade fuse is rated for only 32 V, below a charged 8S pack's 33.6 V.[11] Its 16-week lead time puts it on the first order.

5.0 TEST PLANNING

Tests are ordered by the risk they retire (Table 09); each names its method and verifies the requirements in its SR column; the Fowler test (TC-09, 15 Jan) measures trimmed $C_{L,\max}$ in prototype stalls at a weighed mass and cycles the tracks 100 times, and the tow rig flies 3–8 kg darts. Flights build up from taxi and a trimmed first flight at the lightest loading, through stalls and flap deployment, to payload in steps; a flight is aborted if control or CG margins are off, a temperature exceeds its rating or the pack reaches its reserve. The prototype logs at least 20 flights before freeze, each on a written checklist (§4.2.5 fail-safe, range check, LiPo sacks). Top risks go first (likelihood, consequence): Fowler under-delivers (M, H; TC-09, else the B wing), dart swing (M, H; TC-01), stow failure (H, M; TC-05), fuse opening (L, H; TC-02).

TABLE 09 Test plan (TC-01 to 08 ground, 09 to 14 flight); a failure triggers the Section 3.0 alternative.

TEST, METHOD	SR	PASS CRITERION	DATE
TC-01 Tow rig	08	No divergent swing	Nov
TC-02 Thrust stand	19, 20	$\geq$ 95 N, fuse holds	Nov
TC-03 Drop rig	12	No damage	Dec
TC-04 Sun test	07, 09, 14	Read at 200 m	Dec
TC-05 Drum bench	06	98 % recovery	Dec
TC-06 Tip lift	15–19	§4.2 checks pass	Dec
TC-07 Weigh, CG	01, 05	Mass $\pm$15, CG $\pm$3 %	Jan
TC-08 Timed drills	02, 13	Under 240 s	Jan
TC-09 Fowler test	20	$C_{L,\max} \geq 1.75$	Jan
TC-10 Timed laps	20	Model $\pm$15 %	Jan
TC-11 Stepped M2	03, 04, 21	5 laps < 300 s	Feb
TC-12 Re-check	15–19	§4.2 checks pass	Feb
TC-13 Full M3	06–11	3 modes, stowed	Mar
TC-14 Hot-day run	20, 21	15 % of 97.7 Wh left	Apr

[11] Littelfuse MAXI 32V and MAXI 58V (999 series) datasheets; DigiKey listing 0999080-ZXN, read 29 Oct. 2026.